BUSINESS INTELLIGENCE AND DATA ANALYTICS

University of Petra

Institute of Analytics
Recognized Programme

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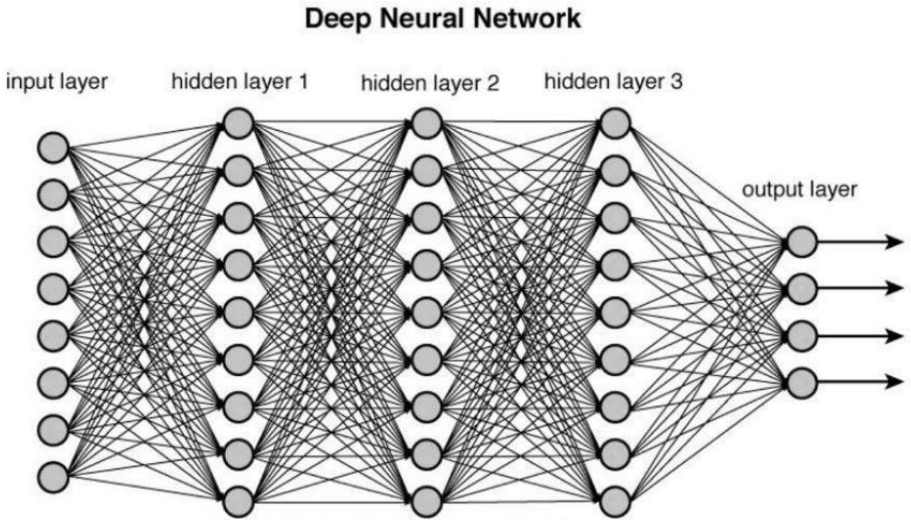
307307
Introduction to Deep Learning and Large Language Models

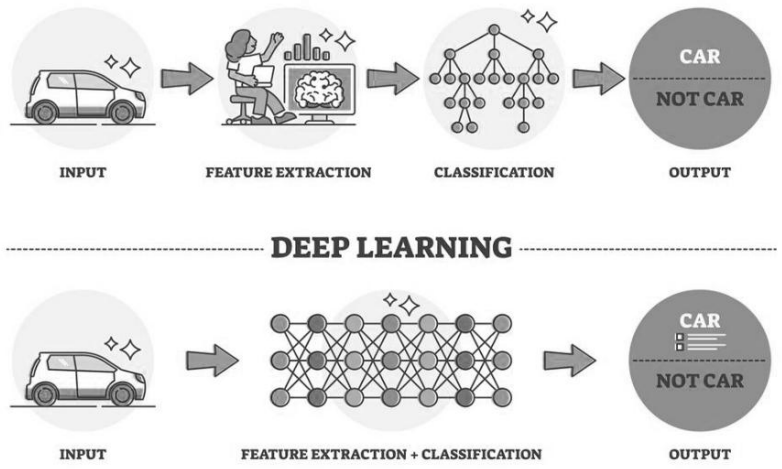
Content

- Introduction to Deep Neural Networks
  - CNNs
  - RNNs
  - The Transformer
  - A Word about Attention Mechanism
- Contextual Word Embeddings
  - Introduction BERT
  - Introduction to HuggingFace and the Transformers Library
  - LLM Training and Fine Tunning

Introduction to Deep Learning

- Neural Networks have revolutionized artificial intelligence by enabling machines to learn from data in ways that mimic human neural processes.
- Deep neural networks (DNNs) are Neural Networks that are composed of multiple processing layers that can learn representations of data with multiple levels of abstraction.
- The power of deep learning comes from its ability to automatically discover complex patterns in raw data through the learning process, without requiring human engineers to manually specify all the knowledge needed by the computer system.





Introduction to Deep Learning

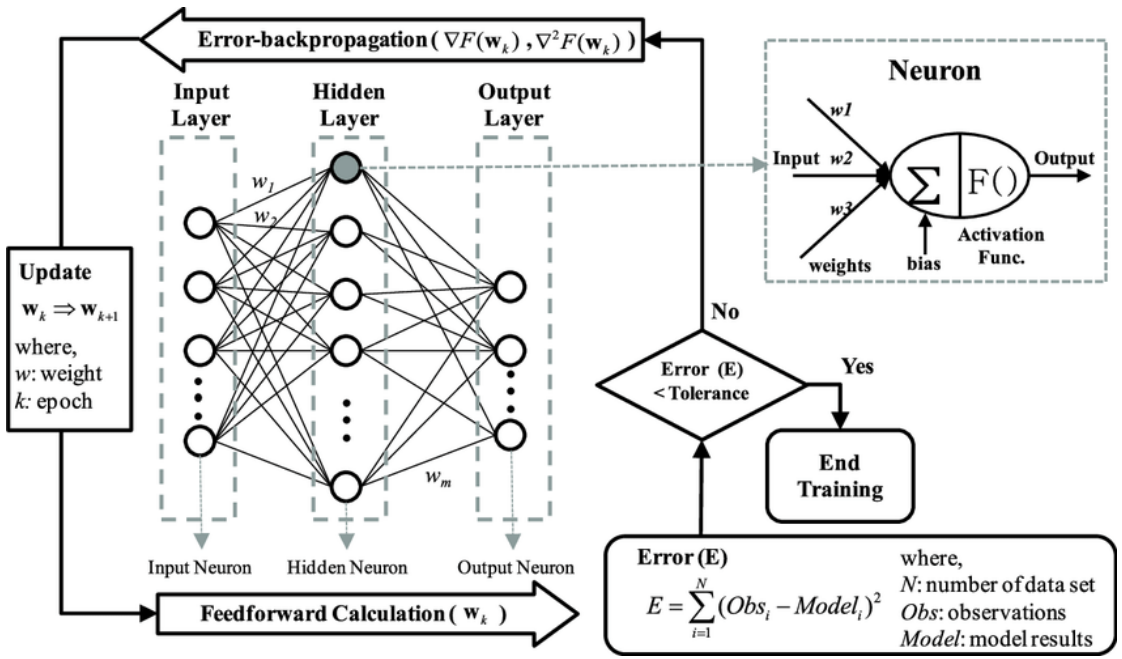
Fundamentals of Neural Networks

At their core, neural networks consist of:

1. **Neurons**: Mathematical functions that take inputs, apply weights, add a bias, and produce an output
2. **Layers**: Collections of neurons that process information in stages
3. **Activation Functions**: Non-linear functions that introduce complexity into the network
4. **Weights and Biases**: Parameters that are adjusted during training

The basic workflow involves:

- Forward propagation: Data flows through the network
- Loss calculation: The network's prediction is compared to the actual value
- Backpropagation: Errors are propagated backward to update weights
- Optimization: Weights are adjusted to minimize errors



Convolutional Neural Networks

CNNs revolutionized image processing by introducing specialized layers that mimic how the visual cortex processes information.

Key components include:

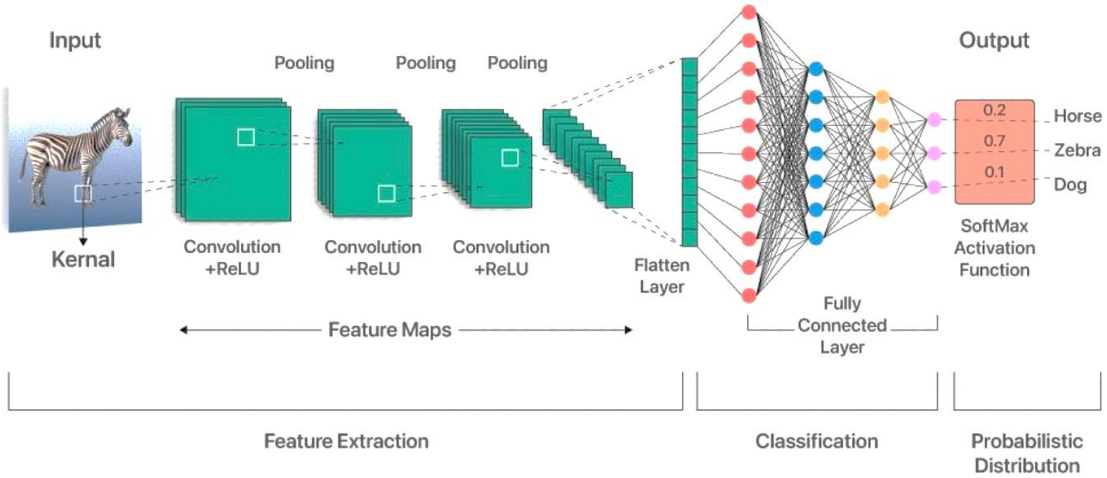
1. **Convolutional Layers**: Apply filters that scan across the input data to detect patterns
2. **Pooling Layers**: Reduce dimensions while preserving important features
3. **Fully Connected Layers**: Connect every neuron to every neuron in adjacent layers

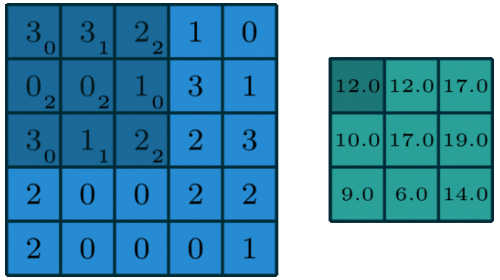
Instead of each neuron connecting to every pixel in an image (which would be computationally expensive), CNNs use:

- **Local connectivity**: Neurons connect only to nearby pixels
- **Parameter sharing**: The same filter is applied across the entire image

Business applications include:

- Product image recognition
- Visual quality control in manufacturing
- Document processing
- Customer behavior analysis in retail





Recurrent Neural Networks (RNNs)

Unlike traditional neural networks, RNNs process sequences by maintaining a form of memory of previous inputs.

Key characteristics:

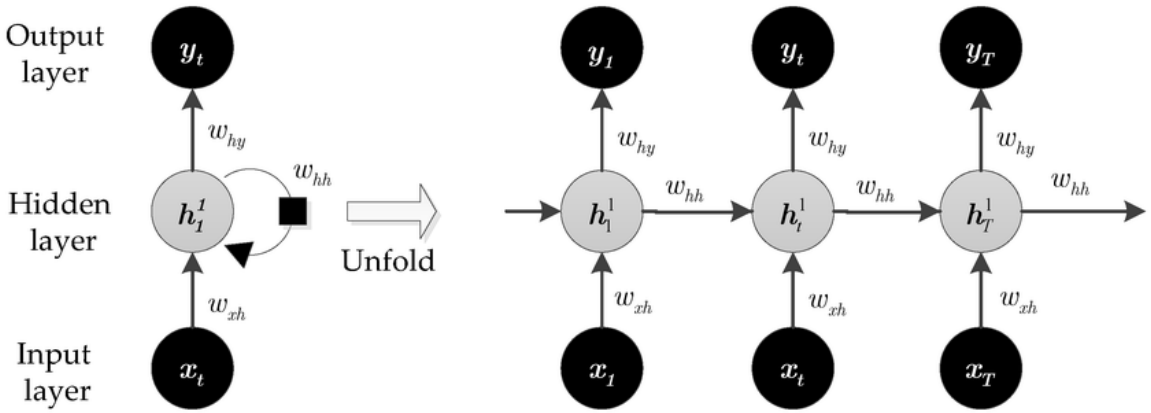
- Time-dependent processing: Output depends on both current and previous inputs
- Shared parameters: The same weights are applied at each time step
- Memory: Internal state acts as a form of short-term memory

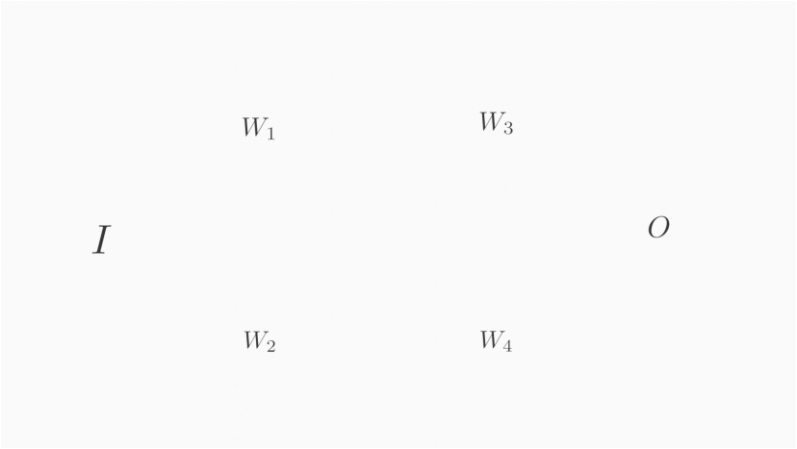
However, basic RNNs struggle with long-term dependencies due to:

- Vanishing gradient problem: The influence of early inputs fades over time
- Exploding gradient problem: Gradients grow uncontrollably during training

Business applications include:

- Language Modeling & Text Generation – Predicting the next word in a sequence (e.g., autocomplete, chatbots).
- Machine Translation – Translating text from one language to another.
- Speech Recognition – Converting spoken language into written text.
- Stock Price Prediction – Predicting future stock or financial data.
- Weather Forecasting – Modeling temporal patterns in weather data.
- Patient Monitoring – Analyzing sequences of medical data (e.g., ECG signals).
- Music Generation – Creating sequences of musical notes.
- Fraud Detection – Detecting unusual sequences in financial transactions.
- Network Intrusion Detection – Monitoring patterns of activity over time.





Transformers

Why move beyond RNNs and CNNs?

- RNNs:
  - Sequential computation → slow
  - Difficulty capturing long-range dependencies
- CNNs:
  - Limited receptive field without deep stacking
- Need:
  - Parallel processing
  - Better long-range modeling

Transformers – Architecture and Principles

- Transformer is a **neural network architecture** based on self-attention mechanism.
- It was introduced in “Attention Is All You Need” (Vaswani et al., 2017), it became the foundation for most modern language models.
- Transformers are optimized to transform a sequence of tokens to another sequence of tokens:
  - Language Translation
  - Q & A Chatbot
  - Text Summarization

Transformers – Architecture and Principles

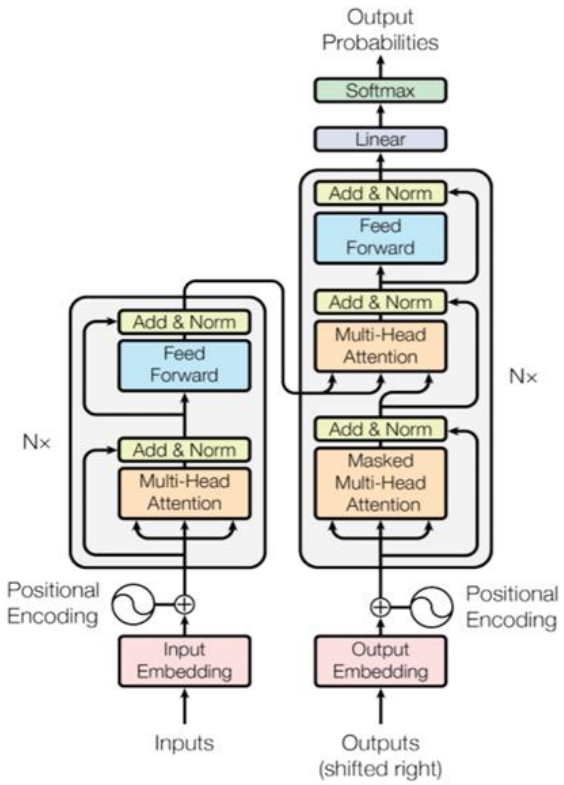
The transformer model consists of two main parts:

1. Encoder:

- The encoder processes the input sequence and generates an encoded representation of it.
- This representation captures the contextual information of the input tokens.

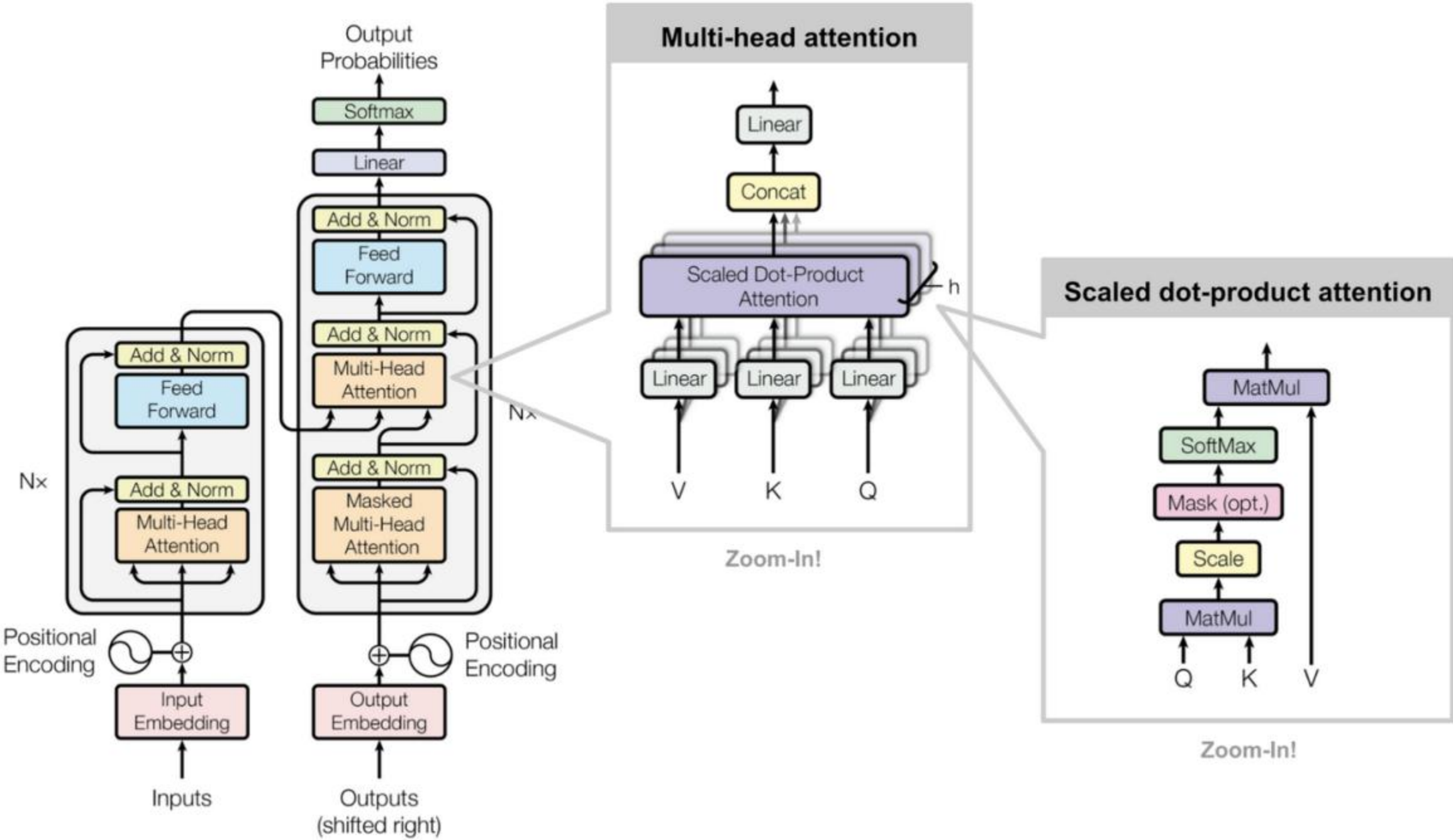
2. Decoder:

- The decoder takes the encoder's output and generates the final output sequence.
- It does this by predicting one token at a time, using the encoded representations and previously generated tokens.



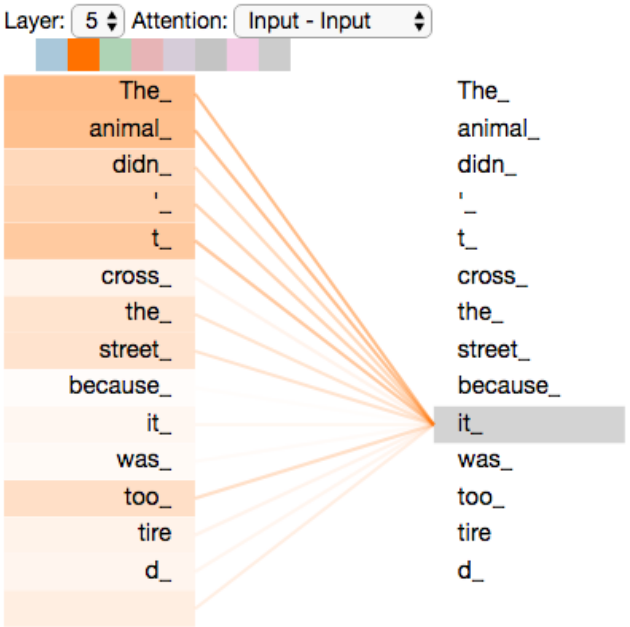
- Encoder: reads and understands text.
- Decoder: generates text one token at a time.

The Self Attention Mechanism


Output Probabilities
Softmax
Linear
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Add & Norm
Masked Multi-Head Attention
Nx
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Nx
Positional Encoding
Positional Encoding
Input Embedding
Output Embedding
Inputs
Outputs (shifted right)
Multi-head attention
Linear
Concat
Scaled Dot-Product Attention
h
Linear
Linear
Linear
V
K
Q
Zoom-In!
Scaled dot-product attention
MatMul
SoftMax
Mask (opt.)
Scale
MatMul
Q
K
V
Zoom-In!

Self-Attention Mechanism

- When the model reads a word, it looks at all the other words around it and gives each one a score—basically asking, *"How helpful is this word for understanding the one I'm working on?"*
- These scores are then turned into weights that tell the model **how much to pay attention** to each word.



The diagram shows which input tokens a specific token is attending to at a given layer and head

Each colored line represents the strength of attention from the selected token (e.g., "it_") to another token in the sequence

Stronger, more saturated lines indicate higher attention weights, revealing where the model is focusing its contextual understanding

	[CLS]	the	cat	chased	the	mouse	[SEP]	it	was	a	fast	chase	[SEP]	[PAD]
[CLS]	0.41	0	0.05	0	0.03	0.06	0.02	0.06	0.15	0.04	0.07	0.02	0.09	0
the	0	0	0	0.01	0.9	0	0.05	0.03	0	0	0	0	0	0
cat	0	0	0	1	0	0	0	0	0	0	0	0	0	0
chased	0	0	0	0	0.98	0	0.01	0.01	0	0	0	0	0	0
the	0.97	0	0	0	0	0	0	0	0.02	0	0	0	0	0
mouse	0	0.04	0	0.78	0.06	0	0.08	0.01	0	0	0	0.01	0	0
[SEP]	0.04	0.08	0.1	0.11	0.03	0.08	0.11	0.06	0.06	0.07	0.05	0.16	0.06	0
it	0.23	0	0.12	0	0	0.17	0.02	0.03	0.14	0.12	0.05	0.06	0.06	0
was	0.13	0.03	0.09	0.02	0.05	0.09	0.08	0.09	0.11	0.07	0.08	0.07	0.09	0
a	0	0.04	0	0.51	0.26	0	0.14	0.02	0	0	0.01	0.01	0.01	0
fast	0	0.09	0	0.67	0.19	0	0.02	0.01	0	0	0.01	0.01	0	0
chase	0.54	0	0	0	0.23	0	0	0.07	0.07	0	0.04	0	0.04	0
[SEP]	0.36	0	0.07	0	0.01	0.1	0.02	0.05	0.16	0.07	0.06	0.02	0.08	0
[PAD]	0.02	0.15	0.03	0.33	0.18	0.02	0.07	0.05	0.02	0.02	0.04	0.04	0.03	0

The heatmap displays attention weights, where each row shows how one token distributes its focus across all other tokens

Brighter cells indicate higher attention values, revealing which tokens are considered most relevant for understanding or generating the current token

Self-Attention Mechanism

Each token produces:

- Query (Q)
- Key (K)
- Value (V)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

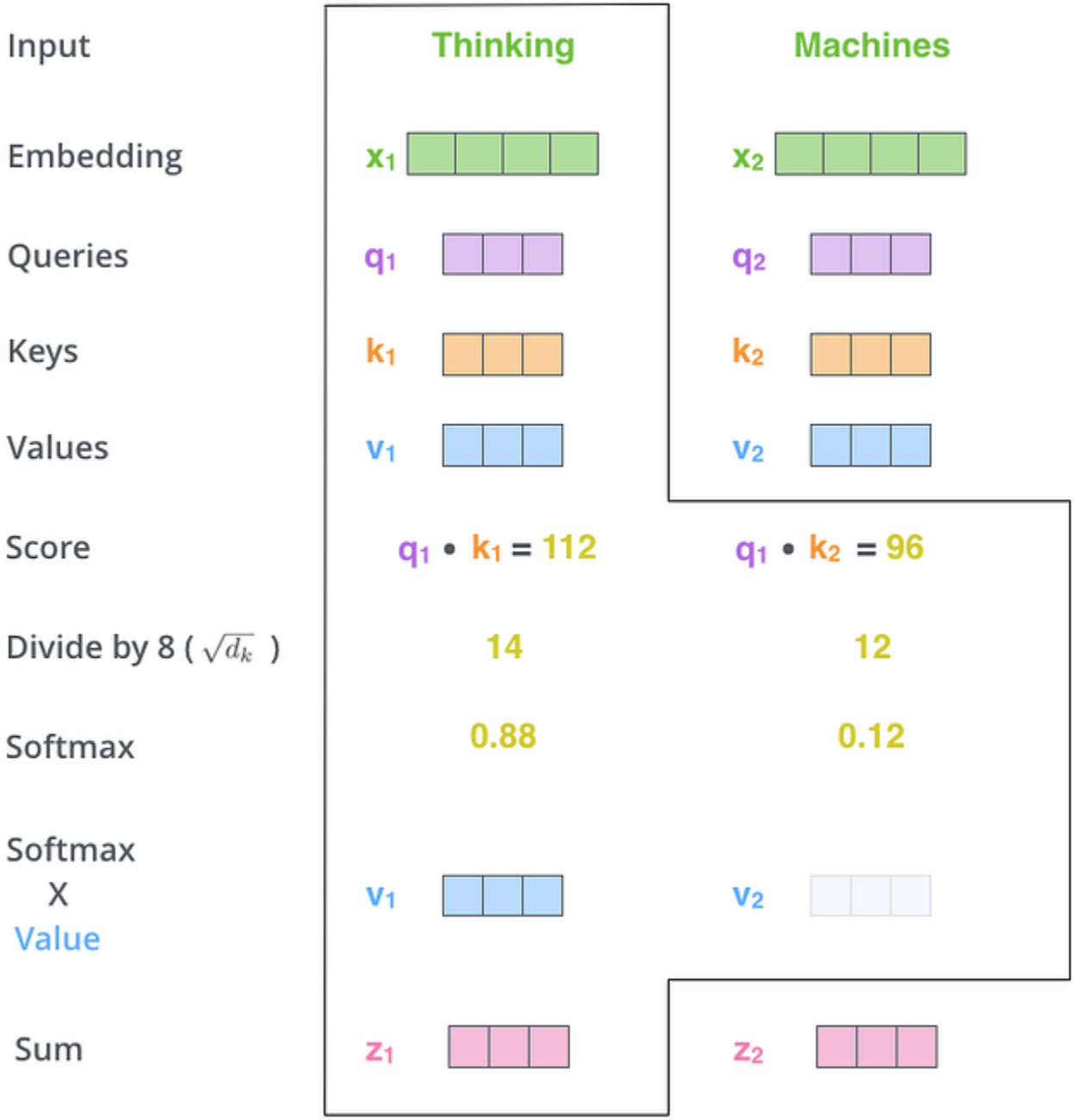
$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

1.Calculate the dot product of the query with all keys to get attention scores: *QK^T*.

2.Scale the scores by dividing by √*dk* (where *dk* is the dimension of the keys) to prevent excessively large dot products.

3.Apply softmax to obtain attention weights that sum to 1.

4.Multiply the attention weights by the values to get a weighted sum.



Simple Attention Example (1 query, 2 keys)

Setup (dimension $d_k = 2$)

$$q = [1,\ 2] \quad k_1 = [2,\ 1],\ k_2 = [1,\ 3] \quad v_1 = [1,\ 0],\ v_2 = [0,\ 2]$$

Step 1 — Dot Products $q \cdot k_i$

$$q \cdot k_1 = (1)(2) + (2)(1) = 4$$

$$q \cdot k_2 = (1)(1) + (2)(3) = 7$$

Step 2 — Scale by $\sqrt{d_k} = \sqrt{2} \approx 1.41$

$$s_1 = 4/1.41 \approx 2.84 \quad s_2 = 7/1.41 \approx 4.96$$

Step 3 — Softmax

$$\text{softmax}(s_1, s_2) = \left[\frac{e^{2.84}}{e^{2.84} + e^{4.96}}, \frac{e^{4.96}}{e^{2.84} + e^{4.96}}\right]$$

Numerically:

$$\approx [0.12,\ 0.88]$$

Step 4 — Weighted Sum of Values

$$z = 0.12 \cdot v_1 + 0.88 \cdot v_2$$

$$= 0.12[1, 0] + 0.88[0, 2]$$

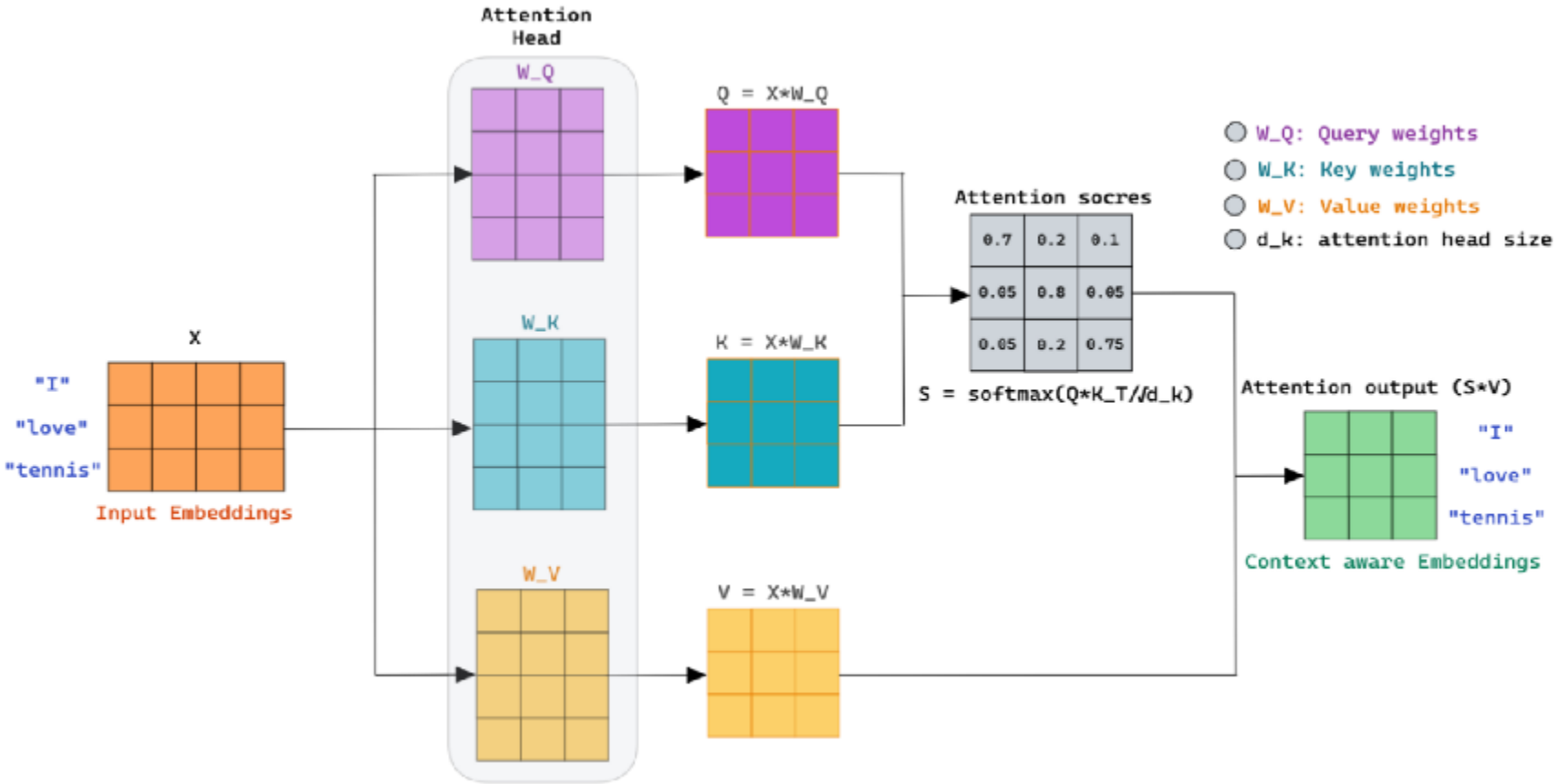
$$= [0.12,\ 1.76]$$

Final Output

$$z = [0.12,\ 1.76]$$

Attention Mechanism

- The diagram shows how AI **pays attention** in a sentence.
- The orange box is the input words (“I love tennis”).
- The middle part is the machine deciding **how strongly each word should pay attention to the others**.
- The green box on the right is the same words, but now each one is **updated with context** (e.g., “love” is tied to “tennis”).



https://www.youtube.com/watch?v=vsElpi4OZWc&list=PLyhJeMedQd9R7u-Dcgh1Zs-DjrAecEPm-

Transformers – The Engine Behind Modern AI

Category	Key Points
How do transformers work?	• Looks at all tokens simultaneously • Uses self-attention to model relationships between tokens • Learns context by understanding how each token influences others
Key building blocks	• Embeddings: convert tokens into numeric vectors • Self-attention: lets tokens influence each other • Feed-forward layers: refine and transform token representations
Why transformers are powerful	• Parallel processing: understand entire sequences simultaneously • Flexible attention: focus on the most relevant information • Scalable architecture: grows effectively with data and compute

How a transformer model completes a sentence?

Step	Stage	Description	Example
01	Tokenization	Input text is split into tokens (words/subwords).	"A puppy is to dog as" → ["A", "puppy", "is", "to", "dog", "as"]
02	Encoding	Tokens are mapped to numerical IDs using the model vocabulary.	["A", "puppy", "is", "to", "dog", "as"] → [101, 2166, 2003, 2000, 2670, 2004]
03	Word Embedding	Token IDs are converted into dense vectors capturing semantic meaning.	101 → [0.12, -0.31, 0.88, ...], 2166 → [0.22, 0.45, -0.12, ...]
04	Decoding	Transformer predicts the next token using self-attention and learned patterns.	Probabilities: "cat" → 0.75, "puppy" → 0.05, "feline" → 0.12 → chosen: "cat"
05	Output	Predicted tokens are converted back into human-readable text.	"A puppy is to dog as kitten is to cat"

Context Aware Embeddings

Contextual Word Embeddings (BERT and GPT)

Key Innovation:

Unlike static embeddings Word2Vec, contextual models generate **different vectors for the same word** depending on its context.

Approach

- Uses deep, pre-trained neural networks (often transformer-based)
- Embeddings are derived from entire sentences, capturing syntax and semantics dynamically

Examples

- **BERT - Bidirectional Encoder Representations from Transformers (2018)**: Transformer-based neural networks trained with masked language modeling and next sentence prediction
- **GPT - Generative Pre-training Transformer (2018):** Transformer-based unidirectional language model focused on generation.

Characteristics

- Embeddings are **context-sensitive** (e.g., “bank” in “river bank” vs. “savings bank”)
- Each word is embedded based on its role in the sentence.
- Embeddings vary for the same word depending on its position and meaning.
- Significantly improve performance on downstream NLP tasks.

BERT: Bidirectional **E**ncoder **R**epresentations from **T**ransformers

- Developed by Google in 2018
- A **pre-trained language model** based on the **Transformer encoder**
- Reads text **bidirectionally**, enabling deep contextual understanding

Key Ideas

- Uses only the **encoder** stack of the Transformer
- Pre-trained on large text corpora, then fine-tuned on specific tasks

Key Advantages

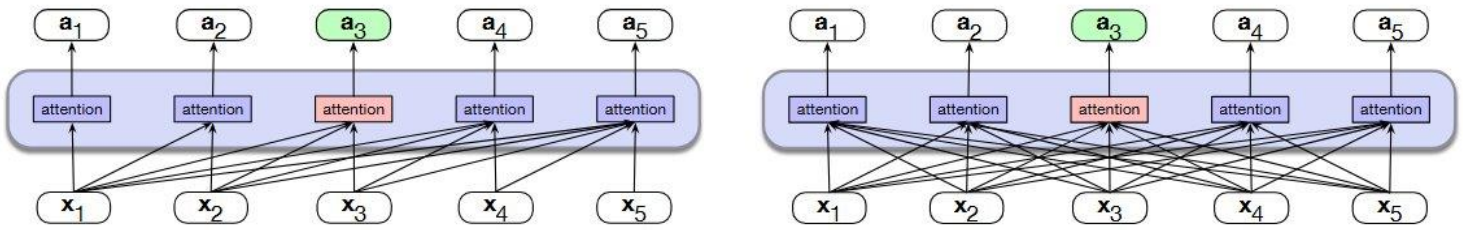
- Contextual embeddings: Word meanings change based on context
- Captures long-range dependencies
- State-of-the-art performance on 11 NLP tasks when released

BERT's Pre-Training Process

- **Training corpus**: 3+ billion words
  - English Wikipedia
  - ~10,000 unpublished books
- **Clean, large-scale dataset** for comprehensive language learning
- **Pre-training objective** to learn language patterns and context

Pretraining Objectives

- **Masked Language Modeling (MLM)**: Predict randomly masked words in a sentence
- **Next Sentence Prediction (NSP)**: Predict if one sentence follows another



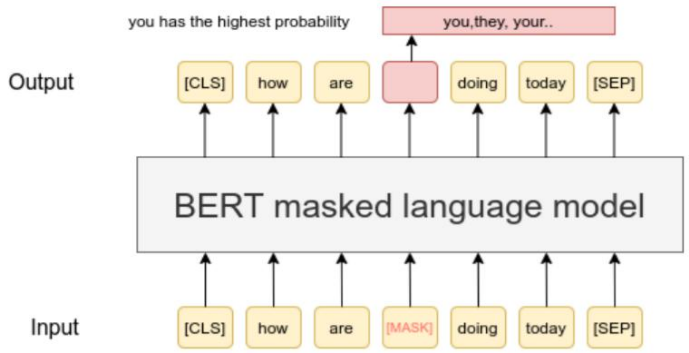
a) A causal self-attention layer

b) A bidirectional self-attention layer

Training Objective 1 - Masked Language Modeling

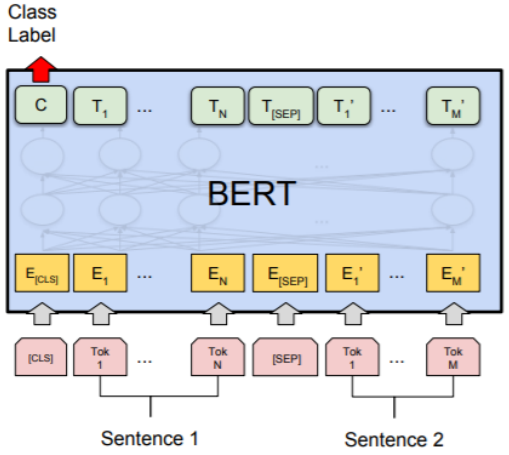
Learning Through "Fill in the Blanks"

- **Random word masking**: Some words are hidden during training
- **Context-based prediction**: BERT predicts masked words using surrounding context
- **Example**: "I love eating _____ in my fruit salad" → "lemons"
- **Key benefit**: Learns word relationships and contextual understanding
- **Powerful concept**: Used in many other AI applications



Training Objective 2 - Next Sentence Prediction

- **Title: Understanding Sentence Relationships**
- **Sentence pair analysis**: Given two sentences, determine if B follows A
- **Logical connection assessment**: Analyzes context and content
- **Example**:
  - A: "The cat climbed the tree"
  - B: "It was trying to catch a bird" ✓ (follows logically)
  - C: "The weather is nice today" ✗ (unrelated)
- **Note**: Later removed in newer models (MLM proved sufficient)



(a) Sentence Pair Classification Tasks: MNLI, QQP, QNLI, STS-B, MRPC, RTE, SWAG

BERT's Ideal Use Cases

Where BERT Excels

- **Text Classification**: Sentiment analysis, topic classification, spam detection
- **Named Entity Recognition**: Identifying people, organizations, locations, dates
- **Extractive Question Answering**: Finding answers within provided context
- **Semantic Similarity**: Measuring similarity between sentences/paragraphs
- **Key advantage**: Perfect for understanding tasks, not generation

GPT – Overview and Architecture

What is Generative **P**re-trained **T**ransformer (GPT)?

- A family of **Transformer-based language models** developed by OpenAI
- Uses only the **decoder stack** of the original Transformer architecture
- Trained with **causal (autoregressive) language modeling** to predict the next token
- Focuses on generating human-like text (unlike BERT's understanding focus)

Training Objective

- Predict the next token in a sequence

GPT Variants

- **GPT-1 (2018, 120M Parameters)**: Introduced the pretrain-then-finetune paradigm
- **GPT-2 (2019, 1.5B Parameters)**: Scaled up model size, trained on web-scale data, Last publicly available weights
- **GPT-3 (2020, 175B Parameters)**: Enabled in-context learning, 100x boost, ~800GB file size
- **GPT-4 (2023, ~1T parameters (estimated))**: Multimodal, stronger reasoning and generalization

Applications (Wide range of NLP tasks through text generation)

- Text generation (e.g., chat, storytelling, code)
- Summarization
- Translation
- Question answering
- Semantic search and reasoning tasks

GPT's Architecture Deep Dive

How GPT Works

- **Multiple stacked decoder layers** with self-attention mechanisms
- **No encoders** - purely focused on text generation
- **Unidirectional processing** - considers only left-side context
- **Next word prediction** based on preceding words
- **High-quality text generation** from learned context patterns
- **Contextual understanding** combined with text creation capability

GPT Training Data

Pre-training Foundation

- **Massive, diverse datasets** including:
  - Web pages
  - Books and articles
  - Billions of words total
- **Different datasets** for each GPT version
- **Large-scale exposure** to language patterns and context
- **Quality varies** but emphasis on diversity and scale

Causal Language Modeling

GPT's Single Training Objective

- **Core task**: Predict the next word in a sequence
- **Method**: Uses context from preceding words only
- **Example**: "I love drinking fresh ___" → "lemonade"
- **Key benefits**:
  - Learns word relationships and context
  - Develops language patterns
  - Enables coherent text generation
- **Simplicity**: No masked language modeling or next sentence prediction

GPT's Ideal Use Cases

Where GPT Excels

- **Text Generation**: Story creation, creative writing, conversations
- **Instruction Following**: Responding to prompts and commands
- **Translation**: Converting text between languages
- **Summarization**: Creating concise summaries of longer texts
- **Code Generation**: Programming assistance and code completion
- **Versatile Applications**: Any task involving text output

Number of Parameters – GPT 3

Component	Parameter	Formula / Size	Total Parameters (Approx.)
Embedding Layer	Token Embeddings	Vocab × d_model = 50K × 12288	614.4M
	Positional Embeddings	n_ctx × d_model = 2048 × 12288	25.2M
Self-Attention (per layer)	Q, K, V, Output	4 × d_model × d_model = 4 × 12288^2	604.6M per layer
Feedforward (per layer)	2 layers (gelu)	d_model × d_ff + d_ff × d_model	1.2B per layer
LayerNorms (per layer)	Two per layer	2 × d_model	24.6K per layer
Total per layer	–	Self-attn + FFN + norms	≈ 1.8B per layer
Transformer Block Total	96 × 1.8B	–	≈ 172.8B
Final LayerNorm	–	d_model	12.3K
Output Layer (tied)	Shared with token embedding	–	— (tied with input embedding)
Total Parameters	–	Sum of above	**≈ 175B**

- **d_model = 12288**
- **n_layers = 96**
- **n_heads = 96 → each head has d_k = d_v = 128**
- **d_ff = 4 × d_model = 49152**
- **Vocabulary size ≈ 50,000**
- **Sequence length (n_ctx) = 2048**

GPT vs BERT Architecture

Aspect	GPT	BERT
Architecture	Decoder-only	Encoder-only
Text Processing	Unidirectional (left-to-right)	Bidirectional
Primary Goal	Text generation	Text understanding
Context	Previous words only	All surrounding words
Use Cases	Generation tasks	Classification/extraction

What is Hugging Face? 🤔

- **A company and a community platform** focused on democratizing Artificial Intelligence, especially Natural Language Processing (NLP) and Machine Learning (ML).
- Often called the "**GitHub for Machine Learning**."
- **Mission**: To make state-of-the-art ML models, datasets, and tools accessible to everyone.
- Started in 2016, initially with a chatbot app, then pivoted to open-source ML.

What does hugging face provide?

1. **Accessibility**: Provides easy access to thousands of pre-trained LLMs.
2. **Standardization**: Offers standardized tools and interfaces for working with different models.
3. **Collaboration**: Fosters a vibrant community for sharing models, datasets, and knowledge.
4. **Innovation**: Accelerates research and development in the LLM field.
5. **Ease of Use**: Simplifies complex ML workflows, from data preparation to model deployment.

Core Components of the Hugging Face Ecosystem

- **Hugging Face Hub**:
  - The central place to find, share, and collaborate on models, datasets, and ML applications (Spaces).
  - Over 1.7 million models, 75,000+ datasets!
- **Transformers Library**:
  - Python library providing thousands of pre-trained models for NLP, Computer Vision, Audio, and more.
  - Supports PyTorch, TensorFlow, and JAX.
  - Makes downloading, training, and using state-of-the-art models incredibly simple.
- **Datasets Library**:
  - Efficiently load and process large datasets.
  - Optimized for speed and memory, built on Apache Arrow.
  - Access to a vast collection of public datasets.
- **Tokenizers Library**:
  - Provides high-performance tokenizers crucial for preparing text data for LLMs.
  - Offers various tokenization algorithms and pre-trained tokenizers.

The Model Hub

Over 1,700,000+ Models Available

Popular Model Categories:

- **Text Generation**: GPT, LLaMA, Mistral, CodeLlama
- **Text Classification**: BERT, RoBERTa, DeBERTa
- **Question Answering**: BERT-based models
- **Translation**: T5, mT5, NLLB
- **Code Generation**: CodeT5, StarCoder
- **Multimodal**: CLIP, BLIP, LLaVA

Getting Started with Hugging Face

- Explore the Hub: huggingface.co
- Browse models, datasets, and Spaces.
- Install Libraries:

```
pip install transformers datasets tokenizers accelerate gradio
```

- Try a Pipeline:

```
# Example: Sentiment Analysis

from transformers import pipeline

classifier = pipeline("sentiment-analysis")

result = classifier("Hugging Face is awesome!")

print(result)

# Example: Text Generation

generator = pipeline("text-generation")

output = generator("In a world of large language models,", max_length=50)

print(output)
```

Under the Hood

1. Automatic model selection
2. Tokenization handled
3. Inference optimization
4. Result formatting
5. Device management

Traditional Approach

1. Load tokenizer
2. Preprocess text
3. Load model
4. Run inference
5. Post-process results

... 50+ lines of code

Other Hugging face Pipelines

The Hugging Face `transformers` library supports a wide range of **pipelines**, each designed for a specific **natural language processing (NLP)** or **vision** task — so you can use powerful models without deep setup.

Pipeline Name	Task Description
"sentiment-analysis"	Classify sentiment (positive/negative)
"text-classification"	General text classification (multi-label or multi-class)
"zero-shot-classification"	Classify into labels **without training** on them
"text-generation"	Generate text (e.g., GPT models)
"text2text-generation"	Text-to-text tasks (e.g., summarization, translation)
"translation"	Translate between languages
"summarization"	Generate a summary of input text
"question-answering"	Extract answer from context
"fill-mask"	Predict missing word in a sentence (BERT-style)
"ner" (Named Entity Recognition)	Extract entities (like names, places, etc.)
"conversational"	Chatbot-style conversation
"sentence-similarity"	Measure similarity between two sentences
"token-classification"	Classify each token (used for NER, POS tagging, etc.)
"feature-extraction"	Extract embeddings/features from a model
"table-question-answering"	QA over structured data (tables)

➤ **Sentiment Analysis**

python Copy Edit

```python
pipeline("sentiment-analysis")("I love this!")
```

➤ **Summarization**

python Copy Edit

```python
pipeline("summarization")("Long article text goes here...")
```

➤ **Translation**

python Copy Edit

```python
pipeline("translation_en_to_fr")("This is amazing.")
```

➤ **Question Answering**

python Copy Edit

```python
qa = pipeline("question-answering")
qa({
    "question": "Where do pandas live?",
    "context": "Pandas are native to China and prefer bamboo forests."
})
```

To list all available pipelines in code:

python Copy Edit

```python
from transformers.pipelines import SUPPORTED_TASKS
print(SUPPORTED_TASKS.keys())
```

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Training LLMs

What is LLM Training?

- LLM Training is the process of teaching a neural network to understand, generate, and manipulate human language.
- It involves feeding the model vast amounts of text data.
- The model learns patterns, grammar, context, and even some level of "knowledge" from this data.
- The goal is to adjust the model's internal parameters (weights and biases) to perform specific language tasks effectively.
- Modern models have BILLIONS of parameters (numbers) to learn during the training process.

Essential Components of LLM Training

1. **Dataset**: Large corpus of text data (e.g., books, articles, websites). Quality, quantity, and diversity are crucial.
2. **Model Architecture**: The neural network structure, predominantly the Transformer architecture (with self-attention mechanisms).
3. **Loss Function**: A function that measures the difference between the model's predictions and the actual target values (e.g., cross-entropy for next-word prediction).
4. **Optimizer**: An algorithm that updates the model's parameters to minimize the loss function (e.g., Adam, SGD).

The General Training Loop

1. **Data Preparation**: Collecting, cleaning, and tokenizing the text data into a format the model can understand.
2. **Model Initialization**: Setting initial random values for the model's parameters.
3. **Forward Pass**: Feeding input data through the model to get predictions.
4. **Loss Calculation**: Comparing predictions to the actual data to quantify error using the loss function.
5. **Backward Pass (Backpropagation)**: Calculating gradients, which indicate how each parameter contributed to the error.
6. **Parameter Update**: Adjusting model parameters using the optimizer in the direction that reduces the loss.
7. **Iteration**: Repeating steps 3-6 for many batches of data over multiple epochs (passes through the entire dataset).

Main Training Phases

1. LLM Pretraining.
2. LLM Fine Tuning

Pre-training: Building the Foundation

- The initial, most resource-intensive training phase.
- Models are trained on massive, diverse datasets (e.g., Common Crawl, Wikipedia, books).
- The objective of this step is to learn general language understanding, grammar, common sense reasoning, and factual knowledge.
- Usually self-supervised (e.g., predicting masked words, next sentence prediction).
- Results in a **base model** with broad capabilities.
- Examples: GPT-3, BERT, Llama pre-training.

Fine-tuning: Specializing the Model

- Takes a pre-trained base model and further trains it on a smaller, task-specific dataset.
- The objective of this step is to adapt the general knowledge of the pre-trained model to perform well on a particular downstream task (e.g., medical question answering, legal document summarization).
- It requires significantly less data and computation than pre-training.
- Can also be used for instruction tuning (following prompts) or aligning with human preferences (RLHF).

Data: The Critical Ingredient

- **Quantity**: LLMs require vast amounts of text to learn effectively. "More data is better" is often true, up to a point.
- **Quality**: Clean, well-formatted, and coherent data leads to better models. Garbage in, garbage out.
- **Diversity**: Exposure to various styles, domains, and perspectives helps create more robust and less biased models.
- **Preprocessing**:
  - **Tokenization**: Breaking text into smaller units (words, sub-words).
  - **Normalization**: Standardizing text (e.g., lowercasing, removing special characters).
  - Creating input IDs, attention masks.

Model Architecture: The Transformer

- The dominant architecture for state-of-the-art LLMs.
- Key Innovations:
  - **Self-Attention Mechanism**: Allows the model to weigh the importance of different words in a sequence when processing information, capturing long-range dependencies.
  - **Positional Encodings**: Injects information about the position of tokens in the sequence.
  - **Encoder-Decoder Structures** (for some tasks) or **Decoder-Only Structures** (common for generation).
  - **Feed-Forward Networks**: Applied independently to each position

Computational Demands & Challenges

- **Hardware**: Requires powerful GPUs (Graphics Processing Units) or TPUs (Tensor Processing Units) for parallel computation.
- **Distributed Training**: Often necessary to train large models across multiple GPUs or machines, adding complexity.
- **Time**: Pre-training can take weeks or months, even with significant computational resources.
- **Cost**: Significant expenses for hardware, cloud computing, and energy consumption.
- **Memory**: Model parameters and activations require substantial memory. Techniques like mixed-precision training help.

Evaluating Trained LLMs

- **Perplexity**: Measures how well a probability model predicts a sample. Lower is better.
- **Task-Specific Metrics**:
  - **BLEU, ROUGE**: For translation and summarization (overlap with reference texts).
  - **Accuracy, F1-score**: For classification tasks (e.g., sentiment analysis).
- **Benchmarks**: Standardized datasets and tasks for comparing models (e.g., GLUE, SuperGLUE, MMLU).
- **Human Evaluation**: Assessing fluency, coherence, helpfulness, and harmlessness by human raters. Often crucial for real-world performance.

Ethical Considerations in Training

- **Bias Amplification**: Models can learn and perpetuate biases present in the training data (e.g., gender, racial, societal biases).
- **Harmful Content Generation**: Potential to generate misinformation, hate speech, or other harmful text.
- **Data Privacy**: Ensuring that sensitive information from training data is not memorized or leaked.
- **Environmental Impact**: Significant energy consumption of training large models.
- **Accessibility and Equity**: Ensuring benefits of LLMs are widely accessible.

LLM Training Details - More Recent Models & Considerations

Model (Version/Size)	Est. Data Size (Tokens)	Context Size (Max Tokens)	Est. GPUs / Compute	Est. Training Time	Est. Electricity / Carbon Footprint	Est. Training Cost
GPT-3 (Base models like Davinci)	~500 Billion - 1 Trillion (incl. C4, Wikipedia, Books, etc.)	2,048 (later models up to 4,096)	~10,000 V100 GPUs (for original run) / ~3.6M A100-equivalent hours (PaLM 540B reference)	~34 days (using 1,024 A100s, research estimate) - Several months (actual, unconfirmed)	~1,287 MWh (training) / ~552 tons CO_2eq (Patterson et al.)	$4.6M - $12M+ (compute, various estimates)
Llama 2 (All sizes)	2 Trillion	4,096	Reported 6,000 GPU-months (A100-80GB equivalent for the family) / 1.7M+ GPU hours for 70B	Jan 2023 - July 2023 (overall project, specific model run time shorter within this)	3.3M kWh (entire project) / 539 tons CO_2eq (training, 100% offset by Meta)	Significant (part of Meta's AI investment)
BLOOM (176B)	366 Billion (1.6 TB)	2,048	384 A100 GPUs	~3.5 - 4 months	~433 MWh (training) / ~25-55 tons CO_2eq (trained in France, low-carbon energy)	~$2M - $5M (compute, public estimates)

LLM Training Details - More Recent Models & Considerations

Model (Version/Size)	Est. Data Size (Tokens)	Context Size (Max Tokens)	Est. GPUs / Compute	Est. Training Time	Est. Electricity / Carbon Footprint	Est. Training Cost
GPT-4	Not officially disclosed (speculated >> GPT-3, likely multi-trillion)	8,192 (GPT-4-8k) & 32,768 (GPT-4-32k); GPT-4 Turbo: 128,000	Not officially disclosed (speculated tens of thousands of A100s/H100s)	~5-6 months (speculative estimates)	Not disclosed (Expected to be significantly higher than GPT-3; estimates range from 20,000-78,000 MWh & thousands of tons CO_2eq for comparable efforts)	Est. >$60M - $100M+ (compute, speculative)
Llama 3 (Instruct models)	>15 Trillion (for the Llama 3 family)	8,192 (some reports suggest up to 128k for future/experimental versions)	Significant clusters of H100s (e.g., Meta mentioned two 24k H100 clusters)	~3 days (8B), ~17 days (70B), ~97 days (est. for 400B+ on 16k H100s)	Llama 3.1 405B est. ~11 GWh. Carbon footprint not yet fully disclosed, but Meta aims for net-zero operations.	Very High (part of Meta's large AI infrastructure investment)
Gemini 1.0 (Pro/Ultra)	Not officially disclosed (multimodal, likely vast & diverse datasets)	32,768 (Gemini 1.0 Pro); Gemini 1.5 Pro: 1 Million (up to 10M experimental)	Trained on Google's TPU v4 and v5e pods (thousands to tens of thousands of TPUs)	Not publicly disclosed (likely months)	Not disclosed. Google emphasizes efficiency & use of renewable energy. Gemini 1.0 was reported to be more efficient than some predecessors.	Very High (part of Google DeepMind's core AI efforts)

Fine-Tuning Large Language Models (LLMs)

What is Fine-Tuning?

- **Pre-training Phase**
  - Trained on large corpus using Masked Language Modeling (MLM) and Next Sentence Prediction (NSP)
- **Fine-Tuning Phase**
  - Fine-tuning is the process of continuing the training of a pretrained LLM on a smaller, task-specific dataset.
  - The objective is to specialize the model for a particular use case or domain.
- **Why Fine Tune LLMs?**
  1. Improve performance on specific tasks.
  2. Inject domain-specific knowledge (legal, medical, financial, etc.).
  3. Adapt to company-specific language or tone.
  4. Reduce inference cost by limiting model size and scope.
- **Use Cases of BERT Fine-Tuning**
  - Customer Support Chatbots (trained on company FAQs).
  - Legal Document Analysis.
  - Scientific Paper Summarization.
  - Code Assistants for specific frameworks.
  - Sentiment classification for product reviews.

Typical LLM Fine Tuning Workflow

1. **Choose a Pre-trained Model:** Pick a base model from Hugging Face (e.g., distilbert-base-uncased for classification or gpt2 for text generation).

2. **Prepare Your Dataset:** Format your data appropriately. Common formats:

- For classification: CSV with text and label columns.
- For generation: Text file or JSON with prompt and completion.

3. **Tokenize the Data:** Convert text into tokens the model understands using a tokenizer.

4. **Define Training Arguments:** Set training parameters like learning rate, batch size, and number of epochs.

5. **Train the Model:** Use a trainer to fine-tune the model on your dataset.

6. **Evaluate & Save:** Check performance and save the model for future use.

7. **Share (Optional):** Push your fine-tuned model back to the Model Hub.

- Create a demo in Hugging Face Spaces.

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